

An Investigation into Elo in Chess and Other Games

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1 Introduction and Background

1.1 Outline

This paper consists of various studies relating to chess and the Elo rating system. We begin with a brief introduction of the Elo system and outlining some mathematics that is common between chapters. During Chapter 2 we will be testing how players can use different time formats to improve their Elo rating. We did this by developing linear regression models to predict rating improvements over 6 months. Chapter 3 will carry out an investigation into a popular topic in chess currently, long win streaks. We investigated whether Elo-derived probabilities can solely explain these long win streaks. Chapter 4 adds to the conversation regarding gender disparity in the sport, using Extreme Value Theory to determine the impact participation rates have on the proportion of female players obtaining the highest ratings in a population. In Chapter 5 we use a set of performance criteria to explore if the Elo system in its current form can be improved and, if so, what adaptations to the current model make this improvement. In Chapter 6 we extend the application of the Elo rating system to team sports, focussing specifically on football. We explore the predictive accuracy of Elo in forecasting league outcomes using Spearman's Rank and Mean Absolute Error.

1.2 History of Elo

The challenge of accurately measuring player skill and tracking performance over time has long been a focal point in competitive games. Before the implementation of modern rating systems, chess rankings were often determined through subjective assessments or simplistic point-based methods, which failed to account for fluctuations in player strength. One of the early structured attempts at a rating system was the Harkness system [1], which introduced an initial framework for numerical rankings but lacked a robust mechanism for continuous skill adjustment. Recognizing these limitations, Hungarian-American physicist and chess master Arpad Elo sought to develop a more statistically rigorous and objective rating system, ultimately revolutionising competitive ranking methodologies [2].

The Elo rating system was formalised in the mid-20th century as a means to provide a mathematically grounded approach to player rankings. In the 1950s, Elo proposed a system based on statistical principles, modelling player performance using a logistic distribution to reflect expected outcomes. His system was officially adopted by the United States Chess Federation (USCF) in 1960 and later by the World Chess Federation (FIDE) in 1970, marking a significant shift in how chess player rankings were determined. Since its adoption, the Elo system has become the global standard for chess ratings and has been extended to other strategic games, including Scrabble, various competitive sports, and online gaming platforms.

1.3 Mathematical Definition of the Elo rating system

At the core of the Elo system is the expected score function, which predicts the probability of a player winning based on their rating difference with an opponent. The system assumes that player strength follows a logistic distribution, ensuring a probabilistic estimation of match outcomes. The expected score E_A for a player with rating R_A against an opponent with rating R_B is given by the equation:

$$E_A = \frac{1}{1 + 10^{(R_B - R_A)/400}} \quad (1)$$

Similarly, the expected score for the opponent is:

$$E_B = 1 - E_A \quad (2)$$

After a game, the new rating of a player is updated using the formula:

$$R'_A = R_A + K(S_A - E_A) \quad (3)$$

where

- R'_A is the new rating of player A,
- S_A is the actual result (1 for a win, 0.5 for a draw, 0 for a loss),
- K is the K-factor, a parameter that controls rating sensitivity to results

In the context of chess, the International Chess Federation (FIDE) typically uses $K = 10$ for elite players, $K = 20$ for standard players, and $K = 40$ for juniors and new players [3].

1.4 Data Background

To investigate the questions in this project, we used various public datasets. To investigate the questions in sections 2, 3 and 5 we utilised datasets from online chess platform Lichess. For section 4, we utilised a dataset directly from FIDE, in particular due to inability of accessing player gender data on Lichess. For section 6, we utilised an English Premier League (EPL) dataset from <https://www.football-data.co.uk/englandm.php>.

Our Lichess data used for section 2 used games from January to June 2017, as any larger period would require infeasible levels of computational power. Over these 6 months over 65 million games were played with over 200,000 registered players. Again limited by computational power, our Lichess dataset for section 3 contained 487,012 games and our dataset for section 5 contained 59,999 games. Both sections 3 and 5 use data from November 2024.

We used Python to preprocess the Lichess data for sections 2, 3 and 5. First we parsed the required information for each game (white player, name, black player name, white Elo rating, black Elo rating, result, and time control) in chunks to handle the large dataset effectively. We proceeded to use the `zstandard` python library to decompress the `.zst` file and saved the required data locally into a new csv ready for analysis. This provided us with a structured foundation for analysis.

We then proceeded to clean the datasets to ensure it contained meaningful and valid data for analysis. First we removed games where either player had rating 1500, the default rating received by new players on Lichess, as these ratings to now represent actual player strength. Further cleaning the data, we removed games that were aborted, unfinished or otherwise had incomplete results. In sections 2, 3 and 5 we reference this process; while we further detail the processing of the datasets in sections 4 and 6 in the relevant chapters.

The dataset used for section 6 covers all EPL matches from 1993 to 2025 and includes information such as the match date, home and away team names, final scores, and season - which was inferred from the date as not explicitly provided. We combined the seasonal CSV files into one unified dataset and then cleaned the data by removing matches with missing or incomplete results. We then loaded the EPL data into a `pandas` DataFrame and sorted it by `Season` and `Date` to ensure chronological consistency for each season. This comprehensive and structured dataset formed the foundation for our analysis of how well the ELO rating system predicts football outcomes.

2 How Novice Players Can Improve Their Elo Rating

2.1 Introduction

In this chapter we will investigate ways that Lichess players of various skill levels can improve their online Elo rating. We will be focusing on how playing different time controls can affect changes in Elo rating over 6 months. The time formats on Lichess are bullet (0-2 minutes), blitz (3-5 minutes), rapid (10-15 minutes) and classical (30-90 minutes). There are countless online discussions between players on how to improve their rating, however responses tend to be mixed, with claims to only anecdotal evidence. Dan Heisman (US National Master) wrote on the issue in 2002 “I always ask, Who are the best fast chess players in the world? Answer: The best slow players”[4]. Heisman attests that players who wish to improve should consistently play many longer games.

While current literature on this matter is limited, 2004 Bruce D. Burns conducted a useful statistical investigation of the subject [5]. Burns first outlines how there are two major skill components in chess: fast processes and slow search. Fast processes include tactical shots or forced checkmates, whereas slow search involves picturing and evaluating variations of future moves. Burns tests two hypotheses over three independent tournaments. Firstly, that a players performance in blitz can account for their variance in rating for other time formats. Secondly, that as a player’s overall skill increases, proportionally more of that skill should be based on fast process. Burns found that performance in blitz chess had correlation between 0.78 and 0.90 with performance in non-blitz chess [5]. This suggests that variance in players’ fast processes accounts for most of the variance in chess skill, and that rating differences were equalised by playing blitz (lower rated players had a better chance of beating higher rated opponents) [5]. Thus, one could advocate that for novice players to improve they should play more blitz games.

In another study, Harreveld investigated the ratings of 300 active online players on the Internet Chess Club (ICC). In direct contradiction to Burn’s results, Harreveld found that skill differences of strong players are to a larger extent based on differences in fast processes than for weak players [6]. Harreveld argues increasing time pressure increases the probability of errors on both sides, resulting in an increase in the absolute number of errors for both strong and weak players. This results in a greater proportional increase in errors for the stronger player [6]. The assumption that elevations in errors are the same regardless of skill seems naive, so to assess this we will conduct a similar investigation on players from Lichess.

2.2 Pre-Processing and Feature Engineering

For this chapter we will be working with the Lichess data set as mentioned in Section 1.4. Before we can conduct our analysis we need to create a data set with the following information: player name, Elo rating at the start of the six months for each time category, Elo rating at the end of the six months for each time category and the number of games for each time format. We made use of Eddie (a Linux computer cluster) to significantly reduce the processing time. From this we can compute the change in rating for each format over the six months. Given that many players tend not to play all time formats, for each change in Elo we have a large spike at zero. Leaving this could cause unexpected problems later, and so before fitting each model we filter for players with more than 30 games who have a positive gain in Elo over the six months.

2.3 Correlation Between Time Formats

In this section we will replicate results discussed in the introduction. To begin, FIDE requires players to reach the ratings detailed in Table 1 before they are given a title.

Title	FIDE Rating
Grandmaster (GM)	2300
International Master (IM)	2200
FIDE Master (FM)	2100

Table 1: FIDE title ratings [7]

Harreveld generated Figure 1 based off players in the ICC. Figure 1 illustrates there is a gradual decline in the correlation between ratings across time formats as Elo rating increases (note that ratings decrease along the x-axis).

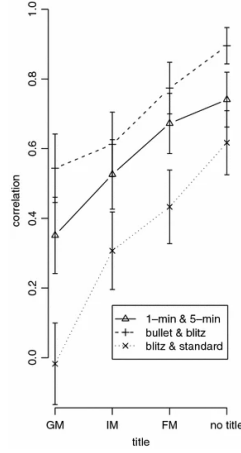


Figure 1: Correlation between Elo ratings - Harreveld [6]

There is some evidence to suggest that lower Elo ratings are less reliable [8]. To combat this, we have already filtered out players who have played at least 30 games. To generate Figure 2, we used the final rapid rating of each player to determine their Elo category. Correlation between all formats reaches a peak between categories 1400-1800 and 1800-2200.

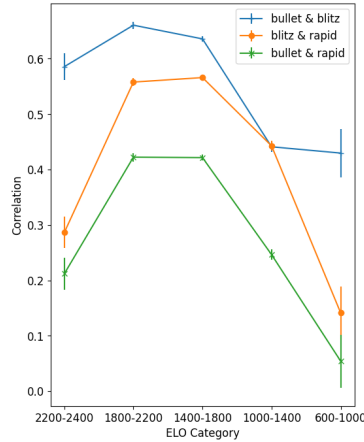


Figure 2: Correlation between Elo ratings

Given FMs have a minimum FIDE rating of 2100, or around 2400 on Lichess [9], Figure 2 supports Harreveld's results for higher ratings. By contrast, for lower ratings we see a clear incline supporting Burn's original claims. This highlights that caution must be taken when making statements about improvement, as findings may differ across the rating ladder. These results do suggest some

pattern across ratings, however, since they are based solely off correlations its is difficult to explain why we see this. More investigation is required to determine the cause of this pattern, which could include looking at the formats of games different Elo categories play or differences in how people learn fast chess compared to slow chess.

2.4 Multiple Linear Regression

2.4.1 Model Formulation

In the next section we will be using multiple linear regression, as outlined in An Introduction to Statistical Learning [10]. This approach allows us to use features to predict the outcome of a response variable. We make the assumption that all predictors are linearly related to the dependant variable, thus the multiple linear regression model takes the form,

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon. [10]$$

where

- Y is the response variable,
- X_j represents the j th predictor,
- β_j quantifies the association between X_j and Y ,
- ϵ is a mean-zero random error term.

It is important to note that the regression coefficients $(\beta_0, \beta_1, \dots, \beta_p)$ are unknown. To estimate the exact model we introduce the parameters $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$. Allowing us to predict our response variable using the formula,

$$\hat{y} = \beta_0 + \beta_1 \hat{x}_1 + \beta_2 \hat{x}_2 + \cdots + \beta_p \hat{x}_p. [10]$$

We estimate the regression coefficients by minimising the sum of squared residuals (RSS),

$$\begin{aligned} \text{RSS} &= \sum_{i=1}^n (y_i - \hat{y}_i)^2 \\ &= \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \hat{\beta}_2 x_{i2} - \cdots - \hat{\beta}_p x_{ip})^2. \end{aligned}$$

2.4.2 Measuring the success of the model

In order to evaluate our model we can make use of various statistics. To begin with, we want to know whether there is a relationship between one of our predictors and the response variable. Using the null and alternative hypotheses,

$$H_0 : \beta_j, \quad H_a : \beta_j \neq 0.$$

In order to test H_0 we calculate the t-statistic for each β_j ,

$$t_j = \frac{\hat{\beta}_j - 0}{\text{SE}(\hat{\beta}_j)},$$

where $\text{SE}(\hat{\beta}_j)$ is the standard error under the assumption the errors (ϵ) are normally distributed (for the number of responses greater than approximately 30)[10]. We define the probability of observing a number greater than or equal to $|t|$ as the p-value. A small p-value suggests that it is unlikely we would observe a noticeable relationship due to chance [10]. To gauge the fit of the models, we could use R^2 ,

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}},$$

where

- $\text{TSS} = \sum (y_i - \bar{y})^2$ and $\bar{y}$ is the mean of the observed data.

In multiple linear regression R^2 is the square of the correlation between the response and the fitted linear model. An R^2 value close to 1 denotes the model's ability to explain a large proportion of the variance in the response variable [10]. One caveat of using R^2 is that with every predictor variable added, R^2 will increase. This is the case even if predictors are not significantly related to the response variable. For this reason we used the adjusted R^2 which accounts for the number of predictors and penalises appropriately,

$$\text{Adjusted } R^2 = 1 - \frac{(1 - R^2)(n - 1)}{n - p - 1}.$$

2.5 Linear Regression Modelling and Results

We utilised linear regression models (outlined in the previous sections) in order to predict changes in Elo ratings. This should inform players of the most efficient ways to improve their rating. First, we grouped players into categories based off their initial Elo. Given that Lichess Elo ratings are inflated¹ we adapted the scale given by A. Elo [2] and we used the categorisation detailed in Table 2.

Rating Range	Category
1999–1800	Class A
1799–1600	Class B
1599–1400	Class C
1399–1200	Class D
1199–1000	Class E

Table 2: Chess Rating Categories

¹A players rating is usually higher on Lichess than other platforms

In order to tune our model, we selected features that are significant in predicting changes in rating, which in this case are features with p-values less than 0.01 (specific p-values are given for each model). We found by removing the constant term (β_0) we could drastically improve our adjusted R-squared value at the cost of an increase in the MSE. When including the constant, we found the model highly weights the constant term. This can be explained by other factors we aren't accounting for such as reading books, watching videos and playing on other sites. Clearly these are important factors and should be investigated, but due to the data we have access to we only used the number of games played. For this reason we decided to remove the constant from our linear regression model.

2.5.1 Rapid Rating

We now run the linear regression model where the response variable is the change in rapid Elo. We find the following results after variable selection,

Rapid Rating Range	Classical Games	Rapid Games	Blitz Games	Bullet Games	Adj. R ²
1800–1999	1.0096	0.0650	0.0967	NA	0.200
1600–1799	0.4523	0.1218	0.1107	0.0654	0.262
1400–1599	0.2429	0.1386	0.0974	0.0763	0.206
1200–1399	0.3892	0.1471	0.1612	NA	0.271
1000–1199	0.4837	0.0961	0.1035	NA	0.210

Table 3: Regression coefficients for rapid Elo gain

Rapid Rating Range	Classical Games	Rapid Games	Blitz Games	Bullet Games
1800–1999	0.000	0.000	0.000	0.068
1600–1799	0.000	0.000	0.000	0.000
1400–1599	0.000	0.000	0.000	0.000
1200–1399	0.000	0.000	0.000	0.029
1000–1199	0.002	0.003	0.001	0.059

Table 4: P-values for rapid Elo gain coefficients

Across all rating ranges we find that classical, rapid and blitz games all have a positive impact on rapid Elo gain. The coefficients represent the expected rapid Elo increase for each game. For example each classical game for Class A players adds 1.0096 rapid Elo rating. For Class A players, classical games are ten times more effective at improving rapid Elo than any other format. As we move down the ratings rapid, blitz and bullet games become more feasible. Notably, bullet games are not significant at changing rapid Elo for Class A, D and E players.

2.5.2 Blitz Rating

Now we will be modelling for the change in blitz Elo. When running the model for all the rating categories we find the following results,

Blitz Rating Range	Classical Games	Rapid Games	Blitz Games	Bullet Games	Adj. R^2
1800–1999	0.8746	0.1624	0.0659	NA	0.194
1600–1799	0.8482	0.1631	0.0732	NA	0.242
1400–1599	0.7385	0.1791	0.1026	0.0517	0.244
1200–1399	0.4859	0.1665	0.1030	NA	0.227
1000–1199	0.6914	0.1981	NA	NA	0.224

Table 5: Regression coefficients for blitz Elo gain

Blitz Rating Range	Classical Games	Rapid Games	Blitz Games	Bullet Games
1800–1999	0.000	0.000	0.000	0.115
1600–1799	0.000	0.000	0.000	0.036
1400–1599	0.000	0.000	0.000	0.001
1200–1399	0.000	0.000	0.000	0.056
1000–1199	0.000	0.000	0.024	0.156

Table 6: P-values for blitz Elo gain coefficients

Similar to the rapid Elo change, Table 5 shows classical games are by far the best way to improve blitz rating. We also see that the other time formats have a positive impact on blitz Elo change and are quite consistent across the rating categories. Contrasting with the rapid model, only we found only one class where the bullet games are statistically significant. Again, the adjusted R-squared value is around 0.2 showing the model is explaining a significant part of the variance in blitz Elo change.

2.5.3 Bullet Rating

Again, changing our model to predict the change in bullet Elo we obtain the following results,

Bullet Rating Range	Classical Games	Rapid Games	Blitz Games	Bullet Games	Adj. R^2
1800–1999	1.9307	0.2265	0.0901	NA	0.252
1600–1799	1.3126	0.1450	0.1575	NA	0.223
1400–1599	0.9656	0.2610	0.0923	0.0539	0.246
1200–1399	0.6507	0.1807	0.1116	NA	0.262
1000–1199	0.4193	0.0881	0.0648	0.1817	0.247

Table 7: Regression coefficients for bullet Elo gain

Bullet Rating Range	Classical Games	Rapid Games	Blitz Games	Bullet Games
1800–1999	0.000	0.000	0.000	0.089
1600–1799	0.000	0.000	0.000	0.256
1400–1599	0.000	0.000	0.000	0.001
1200–1399	0.000	0.000	0.000	0.260
1000–1199	0.000	0.000	0.000	0.000

Table 8: P-values for bullet Elo gain coefficients

Once again, classical games remain the best way to improve bullet Elo gain across all rating ranges. For three of the Classes, the number of bullet games is not a significant predictor of the change in bullet rating. What stands out about Table 7 is for Class E players, playing bullet games is a very feasible way to improve bullet Elo, unlike most other Classes where bullet games are statistically irrelevant. This suggests there might be some time related constraint that Class E players learn best through playing more bullet games. Moreover, we have adjusted R-squared values around 0.2 which is sufficient for our needs.

2.5.4 Optimal way to spend time for rating improvement

One issue with our findings so far is that games from different time formats are not comparable since they last for different durations. We will account for this and investigate the best way a player can spend their time to improve. Based off our previous findings we would suggest players should always play classical games if they want to improve. The problem with this is that many people don't have time to spend 30 minutes on a single game. To account for this in our model we simply multiplied the number games for each format with how long they last. This results in a model which can use time spent in each time format to predict changes in Elo ratings. Within each time control there are multiple formats (eg. blitz includes 3, 3+2, 5 etc.), adapting for this we assume games in each time format lasts the following length: bullet 1.5 minutes, blitz 4 minutes, rapid 12.5 minutes and classical 35 minutes. Running our linear regression models to predict changes in rapid Elo we found the following,

Rapid Rating Range	Classical Time	Rapid Time	Blitz Time	Bullet Time
1800–1999	0.0288	0.0052	0.0242	NA
1600–1799	0.0129	0.0097	0.0277	0.0436
1400–1599	0.0069	0.0111	0.0243	0.0508
1200–1399	0.0111	0.0118	0.0403	NA
1000–1199	0.0138	0.0077	0.0259	NA

Table 9: Adjusted regression coefficients for rapid Elo gain

The coefficients represent the rapid Elo improvement for each minute played of each time category. Contrasting with our games model, we find blitz games are more effective at increasing rapid Elo across most Classes. We do find for Class B and C players playing bullet is by far the most efficient way to improve.

Blitz Rating Range	Classical Time	Rapid Time	Blitz Time	Bullet Time
1800–1999	0.0250	0.0130	0.0165	NA
1600–1799	0.0242	0.0130	0.0183	NA
1400–1599	0.0211	0.0143	0.0257	0.0345
1200–1399	0.0139	0.0133	0.0258	NA
1000–1199	0.0198	0.0158	NA	NA

Table 10: Adjusted regression coefficients for blitz Elo gain

Table 10 shows the coefficients for predicting blitz Elo change. Interestingly, for Classes C and D shorter time controls (blitz and bullet) are best for improvement. Whereas before we found classical games were by far the best way to improve. However for Class A and B players, classical games are the most efficient way to improve. This suggests lower rated have to develop a particular skill before improving, perhaps related to the time restrictions in blitz games.

Bullet Rating Range	Classical Time	Rapid Time	Blitz Time	Bullet Time
1800–1999	0.0552	0.0181	0.0225	NA
1600–1799	0.0375	0.0116	0.0394	NA
1400–1599	0.0276	0.0209	0.0231	0.0359
1200–1399	0.0186	0.0145	0.0279	NA
1000–1199	0.0120	0.0070	0.0162	0.1211

Table 11: Adjusted regression coefficients for bullet Elo gain

Finally, Table 11 details the coefficients for bullet Elo improvement. We found that both classical and blitz games were comparable for bullet improvement. Although we do find blitz games to be more efficient for Class D and E players. Again, for Class A players classical games are still the most efficient way to improve. Now we have accounted for the lengths of games, we found blitz and bullet games are viable ways to improve. Although, even after adjusting for game length, classical games are one of the most efficient formats for improvement across multiple time formats.

2.6 Conclusion and Discussion

In this chapter we have examined how different time formats can affect a player’s improvement. We first extended previous literature to Lichess players, which coincided with both Burns’ and Harreveld’s findings. We then applied linear regression models to investigate how playing different time formats could shape a player’s improvement. Our results indicate that playing classical games is one of the most efficient ways to improve Elo ratings across multiple time formats. Interestingly, classical games become more impactful on a player’s improvement as their rating increases, suggesting that playing long games is vital for improvement at the highest ratings. In contrast, for lower rated players, the most efficient way to improve at blitz and bullet appears to be playing faster games. By using a linear regression model we assume that that learning and Elo improvement is linear, which is a strong claim. To mitigate this we categorised players based off their rating; but a non-linear approach could have been more effective. Additionally, given our adjusted R-squared is around 0.2-0.3, we were only explain 20-30% of changes in ratings. This suggests there are other factors we haven’t considered that could contribute more to a player’s improvement such as, the number of games per day, how long a player analyses their games or how often they complete puzzles.

3 A Study of the Validity of Long Win Streaks in Chess

3.1 Introduction

In this chapter we will investigate long win streaks in chess. Due to the explosion of online chess in recent years there has been a lot of discourse surrounding cheating in chess. Many believe that exceptionally long win streaks indicate cheating. However, this is a misconception rooted in misunderstanding conditional probabilities. While long win streaks are rare, rarity alone does not imply cheating. The probability of cheating given a long win streak is not the same as the probability of a long win streak given cheating. [11] Recently Professor Jeffery Rosenthal of the University of Toronto was recently commissioned by Chess.com to complete a statistical study on the high level chess player Hikaru Nakamura who was accused of cheating when he achieved a 46 game win streak against other high level players. His paper studied the validity of win streaks by Hikaru Nakamura by examining 57,421 of his games to establish a model for the probability of online chess outcomes. His findings suggested that even long streaks could occur naturally within the expected variance of the Elo rating system.

Our investigation will be faced with the same limitations that Professor Rosenthal faced. A player may perform better over a short period due to an increase in study, motivation or positive mental state. With additional complexities arising when we consider the phenomenon of “Boosting”. This is where a rating is artificially inflated, such as by arranging wins in advance with an opponent, or knowingly playing an opponent who is artificially deflating their rating.[12] As well as “Speedruns” - starting an account from an artificially low rating and trying to increase it as quickly as possible [12]. Both “Boosting” and “Speedruns” are against Lichess terms of service and are bannable offences. We need a threshold through which we can decide which streaks to investigate. Therefore we must define what we mean by an “unusual streak”, for our investigation such streaks shall be those of over 50 wins in a row with a draw or loss resulting in the end of a such a streak.

This chapter will ask the question of the existence and validity of “unusual streaks” in chess and if their appearance can be explained by the expected scores from the Elo rating system or if they indicate anomalous behaviour.

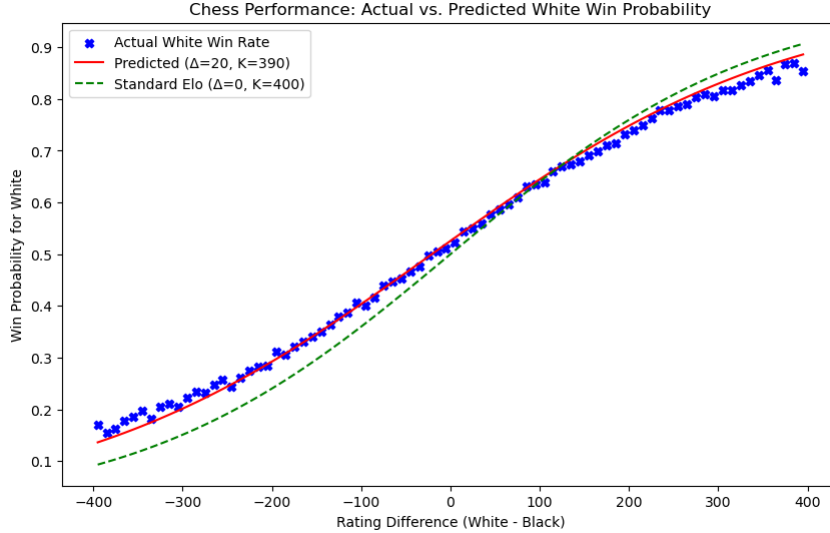
3.2 Processing

In this chapter, we will use a Lichess database. Our chosen file contained 487,012 games. We conducted the process outlined in our data background(Chapter 1.5). However we also needed the longest win streak achieved by every player. Players who haven’t played at least 50 games we taken out of our database as not to truncate longest win streaks. This allowed us to identify those who have

attained an “unusual” win streak. Once those players had been identified we needed certain attributes from every game those players had played detailed in our data background(Chapter 1.5).

3.3 Expected Score

To investigate the probabilities of results in terms of Elo, we will use the observed difference in Elo ratings. Which we can use to obtain an expected score through the Elo formula defined in Chapter 1.3. Using the research conducted by Fecker ,Pires and Bueno Filho[13] on the advantage of starting with the white pieces, we have decided to make a similar change to the standard expected score formula to include a parameter Δ . We began by segmenting the Elo differences into different ranges of 10 eg $(-5, 5), (6, 16)$. Using these segments to calculate the average score for white in our database. This gave us data to fix our Elo-derived win probability. Conducting a mean squared error test on different Δ values against the actual outcomes resulted in us choosing $\Delta \approx 20$ for the best fit. This was to increase the accuracy of our Elo-derived probabilities compared to standard Elo-derived probabilities.



$$E_{\text{white}} = \frac{1}{1 + 10^{(R_{\text{black}} - (R_{\text{white}} + \Delta))/390}}. \quad (4)$$

This formula estimates the expected score for the white player, accounting for the slight statistical advantage of moving first.

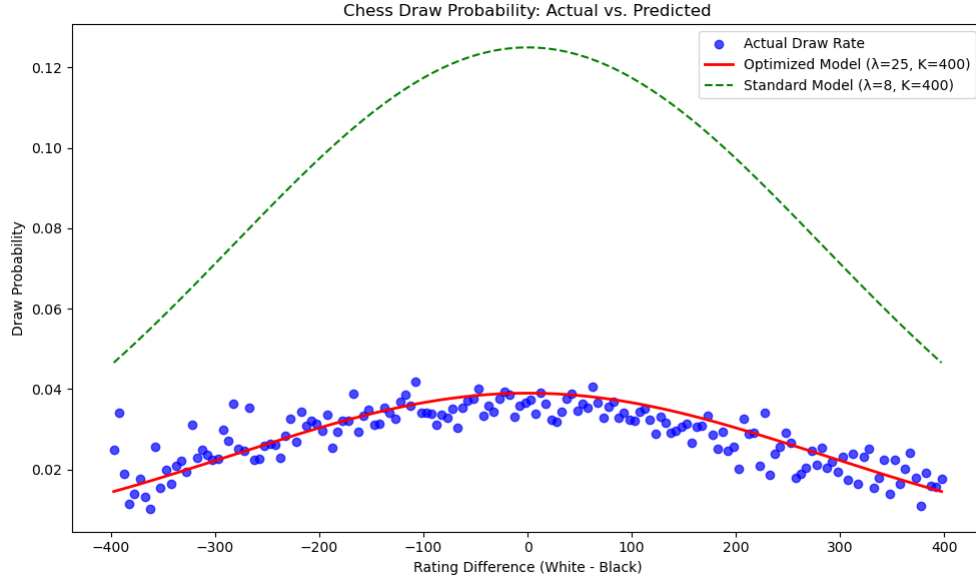
3.3.1 Draws

In chess there is also the outcome that a game will result in a draw. Therefore if we wish to formulate a probability that a player will win a game we must handle the probability of a draw. In our dataset only 4.5% of games resulted in a draw however when we are studying the win streaks we must include the probability of a draw. Using the formula by Professor Rosenthal as our standard formula [11]. We adjusted this formula to best fit our chosen data. This was done by segmenting and testing the data in a similar way as done above. Resulting in the following optimised formula:

$$P(\text{Rosenthal draw}) = \left(\frac{1}{8}\right)e^{-\left(\frac{D}{400}\right)^2}$$

where, $D = R_{\text{white}} - R_{\text{black}}$

$$P(\text{draw}) = \left(\frac{1}{25}\right)e^{-\left(\frac{D}{400}\right)^2}.$$



Therefore we have the probability of a win to be defined as:

$$W = E_{\text{white}} - \frac{1}{2}P(\text{draw}). \quad (5)$$

3.4 “Unusual Streaks”

3.4.1 Identification

We can now identify these “unusual streaks”. We have 4,125 players who have played at least 50 games. While 144 players achieved a longest win streak of

more than 20 games, only 9 players had a longest win streak of over 50 games. The longest win streak achieved was 103 games. Therefore we have 9 individual players who have met our threshold of having an “unusual streak”.

3.4.2 Probabilities

We can now calculate the probabilities based on the observed Elo differences of each player for each of the longest win streaks by using our formula defined above (5). Where the expected score of each game in the streak is multiplied to give a total probability of achieving such a streak.

Player	Games Played	Longest Win Streak	1/prob	Avg. Elo Diff.	Avg. Elo Diff. during streak
houdjini	242	103	77.39	+399	+600
Valance	982	85	637.39	+280	+525
wood manipulator	341	76	518.52	+415	+535
Reppko	1338	64	2453.32	+230	+475
ENERGIE73	1002	62	3473.56	+302	+483
DariusIV	729	50	30.59	+361	+537
Press Resign	90	71	4413.34	+485	+477
monoid	490	55	29.59	+350	+611
dgin666	120	50	345.69	+68	+478

Table 12: The probabilities of the “Unusual Streak”

3.4.3 Analysis

We will begin by conducting some approximation analysis on these streaks where we take the number of games played by each player and divide this up into independent intervals of size the same as the longest win streak that they achieved. We can then approximate the probability of the player achieving such a streak in all of the games that they have played in our chosen month.

$$\text{Adjusted Probability} = 1 - \left[1 - \frac{1}{\text{prob}}\right]^{\text{no. of intervals}}$$

Player	Games Played	Longest Win Streak	1/prob	1/adjusted prob
houdjini	242	103	77.39	38.947
Valance	982	85	637.39	58.40
wood manipulator	341	76	518.52	130.01
Reppko	1338	64	2453.32	123.14
ENERGIE73	1002	62	3473.56	217.57
DariusIV	729	50	30.59	2.69
Press Resign	90	71	4413.34	4413.34
monoid	490	55	29.59	4.16
dgin666	120	50	345.69	173.10

Table 13: “Unusual Streak” Analysis

From this approximation analysis focusing on the two streaks who were the least probable, player “Press Resign” and “ENERGIE73” whose streaks were least

likely with probabilities of 1 in 4413.34 and 1 in 3473.56 respectively. When we construct these independent intervals for these two players to establish a probability of such a streak occurring in the games that they played. We return the approximate probabilities of achieving their respective streaks as 1 in 4413.34 and 1 in 217.57. Considering that there is 4,125 players in our database the the existence of these “unusual streaks” is in line with what we would expect to see as the most improbable streak has a probability of occurring in 1 in 4413. However due to the average Elo differences in the games during each players streak is much higher than the usual Elo difference that players will experience during match making, as Lichess aims to promote fair and balanced games by allowing up to +500 to -500 Elo rating differences with an approximate mean difference of 0 [12]. We can see the Elo differences in the table in section (3.4.2). Alluding to these players purposely choosing weaker players to play against.

3.5 Monte Carlo Simulation

A Monte Carlo simulation is a mathematical technique that simulates the range of possible outcomes for a certain event. Each iteration of the Monte Carlo simulation gets more accurate with each iteration. This property - that the actual probability gets closer to the exact probability with more iterations is known as the law of large numbers.

This technique will be used to conduct further analysis on these streaks than our approximation-styled technique. In our simulations the players who obtained these “unusual streaks” had all of the games that they played in our chosen month simulated 1000 times. Then saving the longest win streak that they achieved in that iteration allowing us to obtain a mean longest streak length that the player would be expected to achieve due to the Elo differences of those games based on our Elo-derived win probability(5). This approach helped account for randomness in performance and provided a clearer understanding of whether observed streaks deviated significantly from expected norms.

3.5.1 Monte Carlo P-Value test

This test allows us to obtain a number that describes the likelihood of obtaining the observed result under a null hypothesis of a test statistic [14]. In our case we performed a p-value test to determine whether observed streaks could be reasonably attributed to Elo-derived expectations. We shall define our p-value test as the following:

$$p = \frac{r + 1}{n + 1}$$

where

- r= number of simulated longest streaks $\geq$ observed longest streak.
- n= total number of simulated results.

To conduct our test, we must establish our hypothesis.

Null hypothesis - “Unusual streaks” can be expected by Elo-derived win probabilities(5).

Streaks with $p \leq 0.05$ were considered statistically significant outliers.

Player	No. of Games	Longest Win Streak	Sim Mean	Sim SD	Monte Carlo P-value
houdjini	242	103	52.00	17.74	0.016
Valance	982	85	41.10	12.85	0.006
wood manipulator	341	76	39.01	12.41	0.013
Reppko	1338	64	54.58	20.62	0.270
ENERGIE73	1002	62	30.66	6.99	0.006
DariusIV	729	50	44.08	12.57	0.289
Press Resign	90	71	31.57	11.68	0.005
monoid	490	55	43.73	11.15	0.156
dgin666	120	50	26.35	9.18	0.005

Table 14: ”Unusual Streaks” Simulation Results

Evaluating the p-values that we received from our test we can see that three streaks fail to reject the null hypothesis, which means that their “unusual streaks” could be reasonably attributed to expected variance in win probabilities. The other six streaks reject the null hypothesis as p-value ≤ 0.05 , suggesting that their performance exceeded expectations based on Elo-derived win probabilities. This leads to the question what factors also played a role in these players achieving these streaks.

3.6 Key Findings

In conclusion, we identified 9 win streaks exceeding 50 games. While the existence of such streaks is not inherently problematic, despite the large difference in Elo between the players. A deeper statistical analysis revealed that 6 of these streaks were not consistent with expected outcomes based on Elo-derived win probabilities. This discrepancy suggests that factors beyond player skill and performance variability contributed to these anomalous streaks.

Specifically, while 3 of the 9 streaks aligned with statistical expectations, the remaining 6 displayed p-values indicating significant deviation from the simulated distributions. These outliers highlight the limitations of the Elo model in accounting for all influences on win streaks, pointing to potential external factors such as opponent selection or psychological influences. We believe that a key factor was non-random matchmaking conducted by players. Account “houdjini” who played the same opponent 60 times in succession which is extremely unusual. Player “dgin666”’s entire win streak was completed against the same player. This type of matchmaking massively contributes to players attaining long win streaks and is very difficult for an Elo model to pick up on. Also from our investigation in win streaks, we believe that further investigation into these anomalous cases could provide valuable insights into chess performance analytics and fair play assessment allowing for improvements to be made to

the Elo rating system to better understand streaks in chess. In an attempt to quantify the physiological influence we would conduct an in person study of players to monitor changes in performance level and what factors influenced this change. All to be used in applications such as finding cheaters and people who are violating terms of service of Lichess to create a better experience for all players.

4 Using Extreme Value Theory to Determine the Impact of Participation Rates on Gender Disparity in High Rated Players

4.1 Introduction

Historically, women have been less successful in the world of competitive chess than men [15–18], with the World Chess Federation listing² 1643 male grandmasters compared to only 37 female [19]. Much research has been done regarding the topic, both analysing statistical factors such as participation rate [16, 17, 20, 21] and psychological factors such as societal gender relations [15, 18, 19]. In this chapter we use extreme value theory to further research on the effect that participation rate has on the lack of extremely high rated female players.

4.2 Background

4.2.1 Motivation

An article by Bilalić et al. published in 2008 [20] analysed player ratings from the German chess federation and argued that 96% of the difference in male and female ratings could be explained by the much smaller sample size of women. This conclusion was reached through analysis that measured the differences between the top 100 male and female ratings. However a follow up article by Knapp published in 2010 [21] pointed out that [20] had made a mistake in its assumption of the data being normally distributed and its model predicted a max male Elo rating of 3031, far above the highest rating recorded of any male player in history (2882, [22, 23]). Knapp attempted to rectify the faulty analysis by fitting the data to a hypergeometric distribution before performing similar difference analysis, coming to the conclusion that sample size difference could only explain around 67% of the difference in extreme ratings.

There have been other papers attempting to measure the effect of participation rates on the abilities of different subsets of players. A paper from 1996 [17] used log-linear analysis to determine that the dominance of men and Russians in high ranks when compared to women and other nationalities could be largely explained by difference in participation rate. Another paper from 2006 [16] compared ratings of male and female players at a variety of age ranges and came to a similar conclusion to [17] through the discovery that male and female players from geographical areas with equal participation rates generally showed no significant difference in skill. Both papers however commented further research was needed on the topic as both had fairly specific parameters of analysis.

We will attempt to comment on the topic regarding gender by using extreme value theory (EVT) analysis, a branch of statistics used to analyse properties

²As of June 2019

of extreme values in data [24].

4.2.2 History of EVT

The study of extreme values can be traced back to the early 18th century with Bernoulli first asking the question: “*n men of equal age die within t years. What is the mean duration of the life of the last survivor?*”. He would later simplify the question, reducing it to asking what the average longest distance from the origin would be for a group of various points on a line [24]. The field would later be formalized in the early 20th century when L.H.C. Tippet was coming up with ways to invent stronger cotton thread. This culminated in the Fisher–Tippett–Gnedenko theorem, which stated that normalised maxima of a sequence of independent and identically distributed (iid) random variables must always be distributed in the form of one of three convergent distributions. These distributions are defined by the following formula (as stated in [25]) for RVs x :

The class of extreme value distributions is $G_\gamma(ax + b)$ with $a > 0$, $b \in \mathbb{R}$ where

$$G_\gamma(x) = \exp\left(-(1 + \gamma x)^{-1/\gamma}\right), \quad 1 + \gamma x > 0$$

with $\gamma \in \mathbb{R}$ and for $\gamma = 0$ the RHS becomes $\exp(-e^{-x})$

The parameter γ is known as the *extreme value index* and defines the distribution type. For $\gamma > 0$ the distribution is known as the Fréchet class of distributions, for $\gamma = 0$ the Gumbel distribution (or the double-exponential) and for $\gamma < 0$ the reverse-Weibull class of distributions.

Commonly, EVT is used in scenarios such as assessing risk in finance, testing engineering limit cases and determining the chances of extreme weather events. One common method is the Block Maxima method [26], which bins data into non-overlapping time periods and examines the most extreme events in each period. This is useful when comparing a small number of values over specific periods of time, such as annual dates of peak rainfall, but is not useful for analysing tail behaviour of Elo distributions. Here we want to analyse large clusters of data points and will use the other common EVT method of Peaks over Threshold. This is a “threshold method” [26], where a value u is selected as the data threshold and the analysis is performed on the data exceeding³ this threshold. For all values $X_i > u$, we turn them into “exceedances” through the transformation $Y_i = X_i - u$. These exceedances are then fitted to a Generalized Pareto Distribution (GPD), defined by parameters ξ (shape), σ (scale), and μ (location). The location of a GPD is simply the value of the smallest exceedance, so as we will only be dealing with the exceedances themselves and have already “shifted” the data by u , our location is 0. Knowing this, [27]

³or preceding if you are examining extremely low values

defines the cumulative distribution function of the GPD as:

$$F_{(\sigma, \xi)}(x) = \begin{cases} 1 - \left(1 + \frac{\xi(x)}{\sigma}\right)^{-1/\xi} & \text{for } \xi \neq 0 \\ 1 - \exp\left(-\frac{x}{\sigma}\right) & \text{for } \xi = 0 \end{cases}$$

The scale parameter is a positive real value that represents the relative spread of exceedance values and the shape parameter is a real value that determines all other characteristics of the tail [26].

4.2.3 Method

A GPD is used to model the tail behaviour of a distribution, and so is useful for comparing the tail behaviour of two different datasets. By fitting our male and female ELO ratings to GPDs we can compare their shape and scale parameters to assess similarity and determine population-irrelevant proportions of extreme value share. We can also perform Monte-Carlo simulations (random sampling of data based on the GPD) to visually compare the distributions.

The threshold value u solely defines the parameters of the GPD due to the parameters fitting around the exceedances [26]. Therefore it is essential to rigorously check that a chosen u is appropriate or else the GPD will not accurately represent the tail. Two plots are often used to visually determine an appropriate threshold, a Mean Residual Life (MRL) plot and a Parameter Stability (PS) plot [26, 28]. An MRL plot is used to identify where exceedances begin behaving like extreme values. By plotting the expected value of the excesses over u (these are the MRLs) against a range of candidate thresholds [29] [30], the plot implies that a GPD is a good fit for the exceedances if there exists a region where the MRL values linearly stabilize as u increases. A threshold candidate u can then be selected from this range because it means the exceedances based on it follow a consistent pattern and therefore a GPD is an appropriate representation of tail behaviour. A PS plot is used to plot ξ and σ parameters of GPDs based on various threshold candidates [26]. This plot is used to confirm threshold regions where ξ and σ are stable and regard these as candidates where a GPD is a suitable fit for exceedance behaviours.

4.2.4 Data

We chose to perform this analysis on the November 2024 FIDE players' rating list, the set of players with ratings obtained from FIDE officiated games. The dataset only includes ratings of at least 1400 in any game mode [31]. This data was appropriate as it tagged players by gender and also listed whether they were inactive⁴. Other available datasets of chess Elo ratings such as from Lichess do not require users to attach a gender to their profile, meaning there is no guarantee that the usable data would reflect the overall behaviour of the

⁴A player is considered inactive if they have played no rated games in the last one-year period [3]

users. FIDE-rated players also represent the most extreme values in the sport, meaning the data is well-suited for extreme value analysis. We will only be analysing the standard ratings of players as this is the most populous rating category.

4.3 Analysis

4.3.1 Data

We first removed inactive players from the data as the inconsistency of when players choose to stop playing, either during their peak or after they’ve dropped in rating, could skew tail behaviour. For example despite being inactive since 2014 Judit Polgar is still rated at 2675, a higher rating than the current⁵ highest-ranked female player Hou Yifan [32]. If a greater percentage of players in one population choose to go inactive at their peak than the other population, this could inflate the extreme values on population and skew our results.

After dropping inactive players and players without a standard rating we had a list of 19777 female ratings and 170741 male ratings, with their highest ratings being 2633 and 2831 respectively.

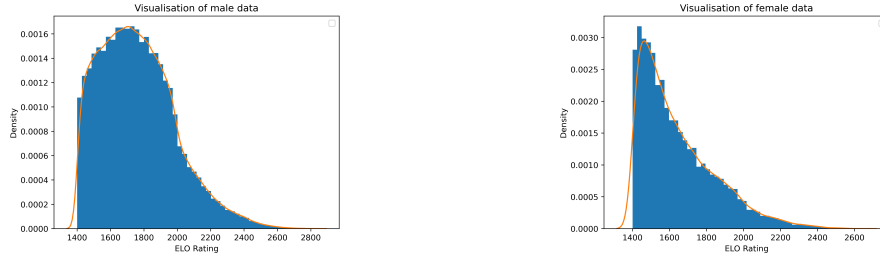


Figure 3: Male and Female data visualisations

Both distributions are right-skewed due to the lower bound of 1400 [31]. The male distribution therefore has a greater spread of values due to its maximum being greater, and we can see that the male distribution has its greatest density at a higher rating than the female distribution. The male distribution has a mean rating of 1761.52 and the female distribution has a mean of 1643.02, matching what we’d expect given the visualisation and historical gender performance in the sport. Both distributions are long-tailed, showing an extremely low proportion of the populations obtain the most extreme values, with the exact proportions of each population obtaining certain extreme values as follows:

⁵as of February 2025

	Male (Empirical)	Female (Empirical)
$x \geq 2300$	2.267%	0.7736%
$x \geq 2400$	0.9998%	0.2376%
$x \geq 2500$	0.3543%	0.05562%
$x \geq 2600$	0.09957%	0.0005056%

Table 15: Percentages of each distribution above certain extreme values

Note that these are the percentages within these datasets where the minimum rating is 1400, and in a population of chess players where the lowest Elo rating was 0 these percentages would be much lower. This should be kept in mind for all further percentages displayed in the chapter. Here we see that even proportionally, men still appear more likely to reach higher ratings than women. The following analysis seeks to determine whether these proportions become more similar if the populations are the same size.

4.3.2 Choosing a threshold

POT method is useful as it allows us to compare the tails of two datasets with different distributions. For our analysis it is most useful to choose the same threshold, this makes it easier to visually compare plots and not need to account for shifts in mean when performing random sampling. However, choosing the same threshold could lead to issues such as including non-extreme values in the dataset with a greater population or producing too little exceedances in the smaller population, and we therefore must be careful to choose a threshold that is not too low for the male population and not too high for the female population.

It seems natural to base our choice of threshold on FIDEs own thresholds for player titles. The lowest title given by FIDE is Candidate Master (CM) which is attained by a rating ≥ 2200 [33]. This threshold gives us 7507 male exceedances and 339 female exceedances, 4.4% and 1.7% of the respective populations. Thus the threshold of 2200 is a good choice of threshold as raising it any higher could mean too few female exceedances and raising it higher could include non-extreme male data points. We must now check whether the exceedances associated are an appropriate fit for a GPD.

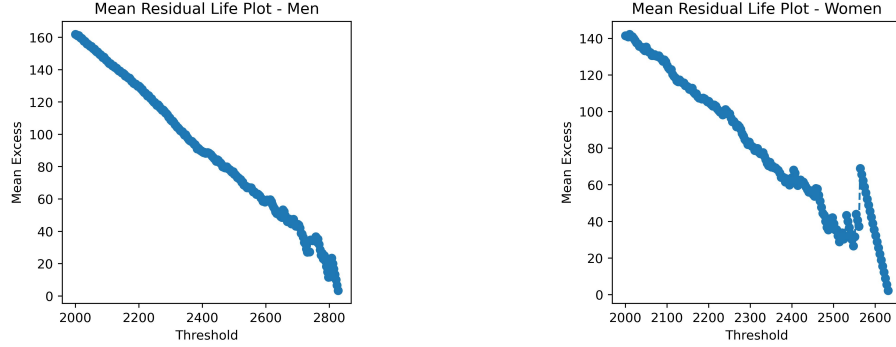


Figure 4: Male and Female MRL plots

Here we see that the MRL values are linear over the threshold range of 2000-2300 in both populations, meaning that 2200 is potentially a suitable choice for fitting a GPD.

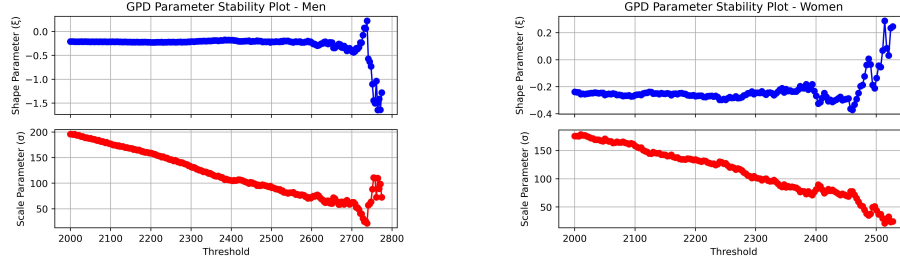


Figure 5: Male and Female PS plots

Both the shape and scale parameters of the GPDs are linear across the threshold range of 2000-2300, consolidating the choice of 2200 as an appropriate value for u . Visual tests have indicated 2200 is an appropriate choice but we must confirm this through goodness-of-fit tests such as performing a Cramér-von Mises (CvM) test or Anderson-Darling (AD) test [34]. Both tests measure how accurately the GPD models the empirical data, with the AD test being a modified CvM test that is weighted more toward assessing tail accuracy. As we are only examining the upper tails and not the distributions as a whole, the Cramér-von Mises test is suitable for measuring goodness-of-fit and is easier to implement. The test runs under the null hypothesis that the empirical cdf and theoretical GPD cdf are the same, with test statistic calculated by the formula

$$\omega^2 = \frac{1}{12n} + \sum_{i=1}^n \left\{ F(X_{(i)}) - \frac{2i-1}{2n} \right\}^2$$

The GPD is a good fit for the data if ω^2 is appropriately low. The *scipy.stats*

package for python contains a function *cramervonmises* [35] that we used to calculate ω^2 and its associated p-value. If the p-value is greater than 0.05 we fail to reject the null hypothesis and the distributions are similar. The test produced a p-value of 0.2588 for the male distribution and a p-value of 0.9325 for the female distribution, meaning we can accept both GPDs as accurately representing their respective tails.

4.3.3 Comparing the GPDs

To visually compare the two GPDs we used random sampling to create simulated datasets. We created a simulated tail sampled from the female GPD with a population of 7507⁶, then compared this to a simulated tail sampled from male GPD that also has 7507 values. Aside from being a useful visual aid, these distributions can be compared using a 2-sample CvM to test for similarity. A 2-sample CvM test [36] compares two empirical distributions for similarity, using a similar process as the 1-sample version. It is appropriate as a comparative test here as once again we are exclusively analysing tails. We used the *stats.genpareto.rvs* function [37] to create the simulated data, setting the location to 2200 for easier visualisation. We ran multiple simulations to account for randomness⁷.

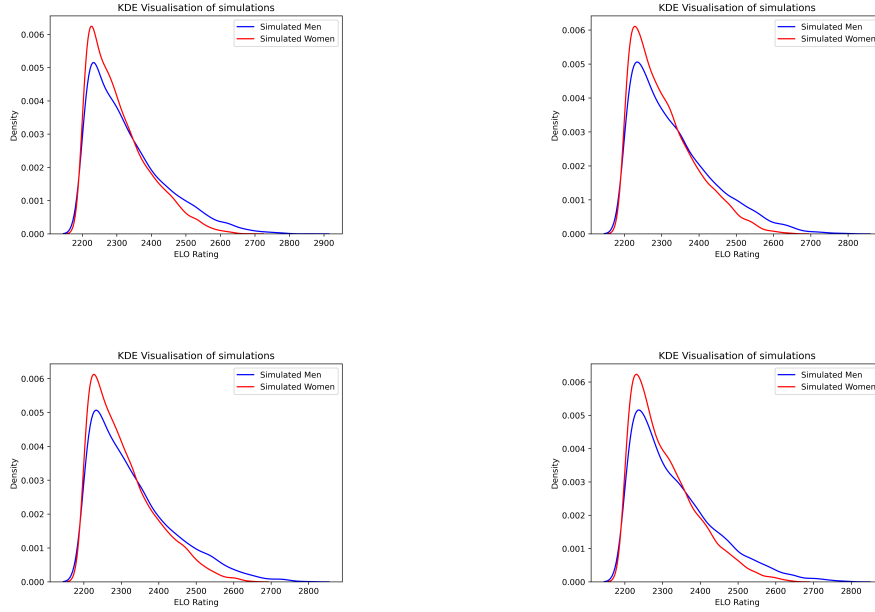


Figure 6: KDE plots comparing male and female simulations

⁶Chosen as this is the number of exceedances in the empirical male distribution

⁷For reproducibility the *random_state* parameter was set to 27, 50, 171, 400 for each respective plot

Visually it is clear that the distributions are not very similar and the CvM test confirms this, rejecting the null hypothesis in each case with p-values to the degree of 10^{-10} .

Using the GPDs we can calculate population-irrelevant probabilities of extreme values in each population using the formula:

$$\begin{aligned}\mathbb{P}(X > \text{rating}) &= \mathbb{P}(X > 2200)\mathbb{P}(X > \text{rating}|x > 2200) \\ &= \mathbb{P}(X > 2200) \left(1 + \xi \frac{x - 2200}{\sigma}\right)^{-1/\xi}\end{aligned}$$

where we have used the survival function of a GPD derived from the cdf. Once again, this is all conditional on $x > 1400$ due to the lower bound on the empirical data.

	Male (Emp.)	Female (Emp.)	Male (GPD)	Female (GPD)
$x \geq 2300$	2.267%	0.7736%	2.2236%	1.9165%
$x \geq 2400$	0.9998%	0.2376%	1.0039%	0.6552%
$x \geq 2500$	0.3543%	0.05562%	0.3763%	0.1433%
$x \geq 2600$	0.09957%	0.0005056%	0.1059%	0.0103%

Table 16: Comparison of empirical extreme rating proportions and those derived from GPDs

We can see the proportions of extreme values have become much more similar, but there is still a large difference as values become more extreme. However the change in proportions are very substantial, especially in the most extreme case where the proportion of women rated above 2600 goes from 0.0005% to 0.01%, showing sample size has a considerable effect on extreme value proportions. It is worth noting that these percentages are also calculated under the assumption that the proportions of the populations rated above $u = 2200$ stays the same as sample size changes, which we cannot assert as true.

4.3.4 Shifting Parameters

Our results so far have indicated that even if sample size does have some effect on the lack of extremely high-rated female players, there still exist other underlying reasons contributing to the discrepancy, especially as values get more extreme. It is worth checking whether a certain parameter is causing most of the difference, as in real life the introduction of a larger sample size could have an effect on parameters that are not represented through repeated sampling.

Bootstrapping [38] (1000 values) to obtain a 95% confidence interval for the shape parameters of each GPD gives us an average shape of -0.2251 with a 95% CI of $(-0.2460, -0.2067)$ for the male distribution and an average shape of -0.2807 with a 95% CI of $(-0.3950, -0.1890)$ for the female distribution.

These confidence intervals overlap so we can assert the shapes are similar, and comparing a simulated distribution based on a GPD defined by the male shape parameter and female scale parameter and the simulation based on the female GPD confirms that the distributions are similar with a 2-sample CvM test returning a p-value of 0.2271 (to 4 s.f.).

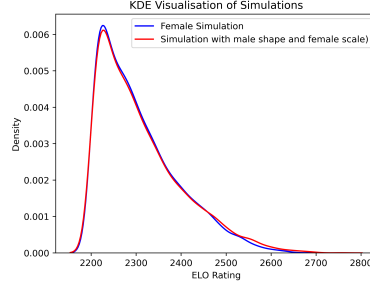


Figure 7: KDE plot comparing male simulation and female simulation using male shape parameter

Performing the same bootstrapping process for the scales of each distribution gives us an average scale value of 158.1223 with a 95% CI of (153.0393, 162.9830) for the male distribution and an average value of 134.3038 with a 95% CI of (115.8444, 153.5951) for the female distribution. These confidence intervals have barely any overlap, implying the scales are very different. This is backed up by figure 7 which showed that the distributions become similar when they have the same scale, and in table 17 where we can see the probabilities of attaining extreme values have become far more similar, beginning to substantially deviate at a greater threshold than when scales were different.

	Male (GPD)	Female (GPD)	Female (GPD-Adjusted Scale)
$x \geq 2300$	2.2236%	1.9165%	2.2213%
$x \geq 2400$	1.0039%	0.6552%	0.9512%
$x \geq 2500$	0.3763%	0.1433%	0.3176%
$x \geq 2600$	0.1059%	0.0103%	0.0662%

Table 17: Comparison of extreme proportions with GPD fit to female shape and male scale

When the scales are identical the male distribution still displays a significantly higher proportion of the highest extreme values, showing there exist some noticeable underlying differences between the populations conveyed by the small difference in shapes.

4.4 Conclusions

Prior research has indicated societal gender relations play a large role in the lack of extremely high-ranking female chess players. One paper [19] relates gender discrepancies in chess to economic gender disparities in various countries around the world, finding that there was a huge variation in proportion between countries. The paper found that socialist countries that promoted greater gender equality tended to have a higher proportion of high-ranking female chess players. Some of these findings are corroborated by the paper [39], which compared gender competitiveness between matrilineal and patriarchal societies, and another paper [18] found that female chess players performed worse when knowing their opponent is a man and often adopted a more defensive playing style, however when not knowing the gender of their opponent they performed fairly equal against men and women, an effect known as “stereotype-threat” that is also referenced in [16].

This specific phenomenon could have a particularly strong effect on FIDE ratings as the ratings are largely determined by results from in-person competitions⁸, and further studies could be done comparing the findings presented in this report to ratings to similar analysis of a population of players who play exclusively online.

Our findings suggest that participation rates do indeed have an impact on the amount of extremely high-ranking players, although as stated in [40] there appear to be other factors at play such as socio-cultural and simply individual differences. Further research should be done on the topic, with a focus on whether or not an increase in population has any effect on the scale of a population’s extreme values, and how GPD parameters vary between in-person rated and online rated populations.

⁸The only way to alter a FIDE rating online is through the FIDE Online Arena server

5 Can The Elo Rating System in Chess be Improved?

5.1 Motivation

5.1.1 What Does It Mean to “Improve” a System?

At first, this may look like a trivial question. We may use words and phrases like “better”, “stronger” or “superior” often without recognising the irony of using synonyms to define our target word.

Upon realising this, this definition becomes much more difficult. Yet, it is integral to the attempted findings of this section, and so our starting point is obvious: to identify, discuss and finalise clear criteria for what an “improved” system looks like in the context of a chess rating system. We define and investigate the following three terms to help us determine “improvement”:

Predictive Accuracy: an “improved” system should predict results more accurately. Our motivation to better this is simple: the closer the gap between predicted and actual results, the more confident we can be about player strength given their rating. This is our primary criterion for improvement, and hence will be weighted through utilisation of two separate metrics to measure predictive accuracy.

Stability: an “improved” system should not overattribute short-term performance. By minimising any unrealistic fluctuations caused due to large swings in short-term performance, we provide a more reliable estimate of player strength.

Interpretability: an “improved” system should be easy to compute by hand, and hence interpretable to the average chess player. We will aim to balance simplicity with predictive accuracy, prioritising the latter if we are made to choose.

We will hereby refer to these as our “improvement criteria”, which will be used as the basis for answering our core question throughout the rest of this section.

5.1.2 Why Might the Elo Rating System Need Improvement?

Historical Context

Elo became the official rating system for The World Chess Federation (FIDE) in 1970 and is still in use today. This begs the question: why even try to improve Elo? Is it even feasible that a system that has stood the test of time, over 50 years, could be improved?

The answer, in our opinion, is yes. Factors like the rise of online chess; increased availability of data from chess tournaments; and access to better tools for analysing that data, all contribute to the fundamental assumption that this section relies upon. That is: finding an alternative, “improved” model for mea-

asuring the strength of a chess is not just realistic, but far more likely today than at any point in the past.

Current limitations

This is not a novel assumption. Over the last 55 years, countless papers have independently researched and identified limitations in the use of Elo, largely intending to facilitate research into new systems that aim to improve Elo. Here we will provide a short overview of those limitations. In an article titled *The rating of chess players past and present* in 1978, Elo himself estimated that, for players of the same strength, the expected score for the player with the white pieces should be 0.57, analogous to roughly a 50-point difference using Elo’s expected score formula. However the Elo rating system does not incorporate this at all, instead estimating 0.50 as the expected score for white in the case of equal player strength. This is a clear example where predictive accuracy can be improved. Research by Fecker, Pires and Bueno Filho [13, 41] referenced in Section 3.3, also looks at this: we will draw upon this research shortly.

Experts have also referenced their dissatisfaction at how the Elo system treats draws in games between higher-rated players [42, 43], where they are generally more common. Many players and analysts believe that at the elite level, the higher-rated player in a game is punished overly harshly for draws.

Research has also highlighted the problems associated with Elo assuming constant player strength and not accounting for variability in ratings. [42] Glickman and Jones explore a demonstrative example with two players who haven’t played in multiple years, with registered ratings 400 apart. The Elo system suggests that the higher-rated player should have an expected score of 0.91; however intuitively we would expect this to be much closer to 0.5. This not only demonstrates the Elo system’s lack of adjustment for recency but suggests an unfairness for players playing unrated or provisionally rated players.

5.2 Background and Literature Review

5.2.1 What Previous Suggestions Have Been Made to Improve Elo?

There has been no shortage of suggestions to improve the Elo system since adoption by FIDE in 1970. Virtually all of these suggestions attempt to tackle one of the limitations referenced earlier.

For example, multiple attempts in the past [13, 41] have introduced the concept of a “white advantage parameter” to integrate this into an “improved” Elo system. Pires and Bueno Filho also introduce the concept of a “drawing propensity parameter” to account for the increased probability of draws in higher-rated games. Incorporating such a parameter could increase predictive accuracy, fairness and scalability, particularly for games between higher-rated

players. Historically however it is the area of variability that has seen the most significant advancements, with changes in this area being officially adopted by FIDE in the past.

One such occurrence was the addition of bonus points and feedback points, introduced in the mid-1970s. This addition attempted to reward the players that “performed exceptionally well” [42] by providing them with bonus points, under the assumption that these were “rapidly improving” players and the typical Elo system did not increase their rating quickly enough. Feedback points were then awarded to their recent opponents to account for the player being stronger than their rating initially suggested. However, these additions were removed from the Elo system that FIDE adopted by the mid-1980s due to lack of statistical backing behind the additions in the first place.

5.2.2 Are There Other Rating Systems to Compare It To?

The most successful attempts to improve on Elo have led to alternative recognised systems, the most notable being the Glicko rating system. Invented by Mark Glickman in 1995, the original Glicko system aims to tackle the challenge of rating reliability often criticised with the Elo system. The Glicko system introduces a “ratings deviation” (hereby “RD”) which measures uncertainty in any player’s rating, where a higher RD indicates a more unreliable rating.

In theory the Glicko system provides better predictive accuracy and stability than the Elo system due to its ability to account for rating uncertainty, while Elo provides a system that is far more interpretable due to its comparative simplicity.

5.3 Developing Solutions

5.3.1 What Alternatives or Modifications Should We Consider?

We will look to remedy three of the most commonly cited shortcomings of the Elo system: the discounting of white advantage, over-penalisation for draws in higher-rated players, and rating unreliability.

To account for white advantage, we will use the previous analysis in Section 3.3, which draws upon recent research to adapt the Elo system to include a white advantage parameter [13] [41], and test the adapted model against the current Elo system across our improvement criteria. We will call this Model A.

Further to reduce penalisation for draws in higher-rated players, we will draw upon research to adapt Elo to include a drawing propensity parameter [13] and again test the adapted model against the current Elo system. We will call this Model B.

Finally to investigate potential improvements in rating reliability, we will utilise dynamic K-factors to adapt the Elo system to adjust rating changes based on rating difference, again testing against the current Elo system across improvement criteria. We will call this Model C.

5.3.2 The Approach

To measure the success of an improvement we will first define key metrics for each of our improvement criteria (two for predictive accuracy, one for stability and one for interpretability).

We will utilise the following performance metrics for each improvement criteria:

Predictive Accuracy

Mean Absolute Error (MAE): Measures the average magnitude of errors between actual game outcomes and the expected scores from the model (predicted results). The lower the MAE value, the better the predictive accuracy of the system. The formula for Mean Absolute Error (MAE) is defined as:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |\text{Predicted}_i - \text{Actual}_i|$$

Where:

- N = Total number of games.
- Predicted_i = Expected score from Elo formula.
- Actual_i = Actual game result (1 for White win, 0 for Black win, 0.5 for a draw).

Log Loss: Measures the confidence in our outcomes (accuracy of the predicted probabilities). Useful for measuring how well a system predicts probabilities rather than just win/loss outcomes. The lower the log loss, the more confident and accurate predictions from the system are. The formula for Log Loss is defined as:

$$\text{Log Loss} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

Where:

- y_i : Actual outcome (1 for a win, 0 for a loss).
- p_i : Predicted probability of a win.

Stability

Stability Score: Measures the standard deviation (volatility) of rating changes to indicate how stable a player's rating on a game-by-game basis. The lower the stability score, the less reactive ratings are to single game outcomes. The formula for Stability Score is defined as:

Volatility_{*j*} = Standard Deviation of Rating Change for
the Player on Average for a Single Game *j*

$$\text{Stability Score} = \frac{1}{M} \sum_{j=1}^M \text{Volatility}_j$$

Where:

- *M*: Total number of players.

Note that in the above formula we assume stability is constant over time. While we note this is a simplification (stability may in fact differ over different time windows) we make this assumption for the purpose of relative comparison between models. Since our analysis is based on the same dataset across the same time period for all models, this assumption should not affect the final conclusion.

Interpretability

Model Complexity: Unit-less metric to measure how simple or complex it is to compute a rating change manually using the system. The lower the model complexity, the more interpretable the model is. We will define this as:

$$\text{Model Complexity} = \text{No. of parameters} + \text{No. of mathematical operations}$$

We recognise both that the choice to minimise the number of operations is somewhat subjective, and acknowledge that counting operations is somewhat arbitrary: different formulations can lead to similar outcomes using a different breakdown of arithmetic steps. However, we argue that this metric remains relevant; in our eyes while the exact number of operations may be arbitrary, this measure reflects the cognitive load required for a user to manually compute the rating calculations. Across all models we also aimed to minimise the number of operations used to compute the rating calculations, allowing us to compare their relative interpretability: the primary purpose of this chapter.

First we will evaluate these metrics for the current Elo rating system and then for Models A, B and C to investigate how the systems compare across our improvement criteria.

A “successful” improvement will be found by comparing our improvement criteria between the Models and Elo. Note that we do not aim to provide an optimal system, if such a system even exists, but instead we simply aim to answer the question of if the current Elo system can be improved, even marginally.

5.4 Implementation and Results

5.4.1 Processing the Data

Referenced in Section 1.3 we utilised a Lichess dataset of 59,999 games to investigate the above. first downloaded a public dataset of the first 59,999 games in November 2024 from online chess platform Lichess in compressed .zst PGN format. We processed and cleaned the data in the manner discussed in Section 1.3.

This cleaning process removed 412 games from our dataset, with the cleaned data of 59,587 games being saved locally as a new csv. This process ensured that Elo calculations are based on games with valid results and player ratings (for testing predictive accuracy), removed games that could lead to bias (for testing fairness), and prepared a clean dataset with required data for subset analysis (for testing scalability).

5.4.2 Implementation for Elo System

Now calculating our performance metrics for the Elo system based on the cleaned data, we find:

Predictive Accuracy

To compute MAE for the Elo system, we used the Elo formula to calculate expected scores for each game, converted actual game results into numeric form and then computed prediction errors. We then proceeded to use the log loss formula above to find the performance metrics:

Mean Absolute Error (MAE): 0.4648

Log Loss: 0.6819

These metrics provide us with a quantitative foundation to compare predictive accuracy between Elo and other systems. We naturally may question such a high value for MAE, particularly given a simple naive strategy of always predicting a draw would provide us with a maximum MAE of 0.5. However, we note that in our dataset draws only occur 3.71% of the time, likely a byproduct of utilising data from an online chess platform and an assumption that is discussed at the end of the chapter. From this we can infer a high proportion of games are decisive: a reason we may see such a high value for MAE. Importantly even if the naive approach were close to Elo’s MAE, the fact each model utilises the same dataset ensures our conclusion comparing different rating models remains

valid.

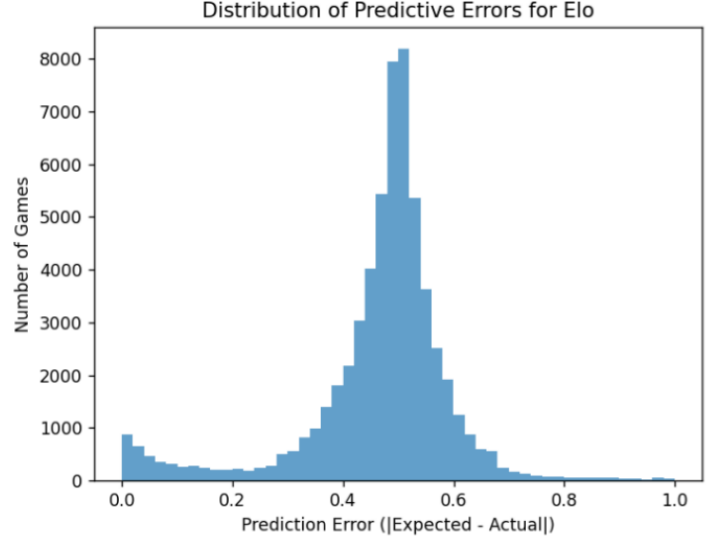


Figure 8: Distribution of Predictive Errors for Elo.

Stability

To compute Stability Score for the Elo system, we first need to choose a K-factor to allow us to calculate new ratings for both players after a game has been played. In line with the current Elo system, we choose the dynamic K-factor of 20 for players with a rating under 2400 and 10 for players with a rating of 2400 or higher.

We then calculated new ratings for both players after each game considering the expected probabilities and actual game results, tracked and stored the magnitude of rating changes for each player and calculated the standard deviation of rating changes (i.e. the stability score). We find:

Stability Score: 3.4522

This provides us with an initial measure of Elo's reliability in maintaining consistent ratings, to be compared to other systems shortly. From Figure 2, we note that the vast majority of players have extremely low volatility in ratings (close to 0.0), which we should expect given that most players consistently play at their pre-established rating level.

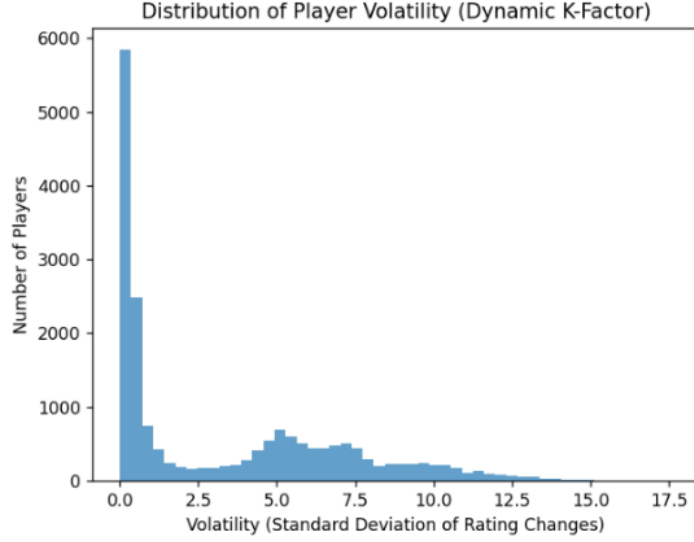


Figure 9: Distribution of Player Volatility

Interpretability

We now use our defined Model Complexity metric to set a quantitative baseline for how complex the Elo system is. We note that the Elo system only has one parameter (namely, the K-factor when evaluating rating change). Breaking down the Elo formula, we find it takes 5 operations to compute the expected score for any particular game and 3 additional operations to evaluate the updated rating for the player. Hence our baseline Model Complexity for the Elo System is $1 + (5 + 3) = 9$.

5.4.3 Implementation for Other Models

To define Model A we will draw upon the previously mentioned research by Pires, Bueno Filho and Fecker. In his research, Fecker analyses all 12,603,842 games recorded in the FIDE database between January 2010 and April 2023 to find that white scores 52.86% of all points in general play. To account for this, we now implement parameter Δ , the white advantage parameter, into the Elo formula as below:

$$E_{\text{white}} = \frac{1}{1 + 10^{(R_{\text{black}} - R_{\text{white}} + \Delta)/400}}$$

We can infer from Fecker’s analysis that when $R_{\text{black}} = R_{\text{white}}$, we have $E_{\text{white}} = 0.5286$. Then solving for Δ , we find that $\Delta \approx 20$. This fits our dataset very closely: by searching for the best value Δ using MAE and checking real game outcomes at near-rating parity, we find that $E_{\text{white}} = 0.5284$ for our dataset and similarly $\Delta \approx 20$. We adapt the current Elo system with this adapted formula for expected score to define Model A, keeping the rating update formula the same.

To define Model B, we will use research from Pires and Bueno Filho to implement a drawing propensity parameter to account for the increased probability of draws in higher-rated games. We introduce P_{draw} to the rating change formula, which scales the magnitude of the rating change based on the likelihood of a draw. Our updated formula is:

$$\Delta R = K \cdot (S - E) \cdot \sqrt{P_{\text{draw}}}$$

Previous research shows we can use a formula based on rating differences to calculate the probability of a draw in the Elo system [44]. This is a Gaussian function where smaller rating differences lead to a higher probability of draw:

$$P(\text{draw elo}) = \frac{1}{\sqrt{2\pi e}} \exp\left(-\frac{\left(\frac{dr}{200}\right)^2}{2e^2}\right)$$

where dr represents the Elo difference between two players.

We then adapted this formula to account for the increased propensity for draws at higher levels of play. To do this, we introduced a scaling factor based on the Elo of the players for each game: utilised to increase P_{draw} for games with higher average Elo. We calculated our scaling factor using the following formula:

$$\text{scaling_factor} = 1 + \alpha \cdot \frac{\mu - M}{M}$$

where:

- μ represents the average Elo rating.
- M represents the median Elo rating.
- α is a scaling coefficient.

We calculated α above to be 0.03163 based on our dataset. We clamped this scaling factor to a reasonable range between 0.5 and 1.5 before adjusting the formula from Xiong et al. by multiplying by this scaling factor, with our final formula being:

$$P(\text{draw elo}) = \left(1 + \alpha \cdot \frac{\mu - M}{M}\right) \cdot \frac{1}{\sqrt{2\pi e}} \exp\left(-\frac{\left(\frac{dr}{200}\right)^2}{2e^2}\right)$$

We adapt the current Elo system with these adapted formulas to account for draw propensity based on rating difference and current Elo to define Model B.

Finally we will define Model C by adapting the Elo system to include dynamic K-factors in the rating change formula. Our dynamic K-factor will be based on rating difference with the following formula for K:

$$K = \frac{40}{1 + e^{|\Delta R|/400}}$$

Now we find the performance metrics across each of the three models to start to build a conclusion on if an improved model has been found.

Predictive Accuracy

Earlier we found the Mean Absolute Error (MAE) of the dataset using the Elo system to be 0.4648. Proceeding to carry out the same calculation on the dataset for the alternative models, we find the Mean Absolute Errors of Models A, B and C to be 0.4539, 0.4603 and 0.4533 respectively.

Similarly we found the Log Loss of the dataset using the Elo system to be 0.6819. The Log Loss metrics for the dataset using Models A, B and C are 0.6664, 0.6776 and 0.6644 respectively.

Given that both the MAE and Log Loss metrics are lower using the alternative models than the Elo system, the above allows us to conclude that all models lead to improved predictive accuracy than the current Elo rating system. We also infer that Model C has the best predictive accuracy of all the models analysed with both the lowest MAE and Log Loss metrics.

Stability

Noting the Stability Score of 3.4522 calculated for the dataset using the current Elo system, we proceed by calculating the stability score for Models A, B and C. We find this to be 3.4988 for Model A, 1.2931 for Model B and 3.2637 for Model C.

Again comparing the data, we find that Models B and C provide lower Stability Scores and hence improved stability to the current Elo system, while the modification made to define Model A provided a higher Stability Score and hence impaired the stability of the system.

Interpretability

Finally we utilise our pre-defined Model Complexity metric to compare the interpretability of the systems. Our Model Complexity for the Elo system is 9.

We adapted the Elo system to include one additional operation and one additional parameter to form Model A, and hence we find the Model Complexity of Model A is 11.

We adapted the Elo system to include seventeen additional operations and no additional parameters to form Model B, and hence find the Model Complexity of Model B to be 26.

Finally we adapted the Elo system to include four additional operations and no additional parameters for form Model C, finding the Model Complexity of Model C to be 13. We conclude that Models A, B and C are all less interpretable than the current Elo system with Model B as the least interpretable and Model A as the most interpretable.

5.4.4 Key Findings

We can summarise our performance metrics in Table 17.

Criteria	Metric	Elo	Model A	Model B	Model C
Predictive Accuracy	MAE	0.4648	0.4539	0.4603	0.4533
	Log Loss	0.6819	0.6664	0.6776	0.6644
Stability	Stability Score	3.4522	3.4988	1.2931	3.2637
Interpretability	Model Complexity	9	11	26	13

Table 18: Comparison of Elo and different models across various criteria.

We can now compare each of the models in turn to the current Elo system. We find that Models A, B and C all provide better predictive accuracy than Elo using both Mean Absolute Error and Log Loss metrics. Models B and C also provide better stability than Elo through our Stability Score metric; however Model A falls short of improving on Elo on this criteria. Finally we also note that Elo is the most interpretable of the systems through our Model Complexity metric, followed in order by Model A, Model C and finally Model B.

The above analysis shows that all Models A, B and C improve on the Elo system across our predictive accuracy criteria; Models B and C improve on Elo on our stability criteria; while the Elo system is more interpretable than any of the models.

In practice, we have shown that either the introduction of a drawing propensity parameter (Model B) or including dynamic K-factors based on rating difference of the players (Model C) would improve the Elo system across key predictive

accuracy and stability criteria, only at the expense of added complexity.

Looking at the history of Elo, we note that improvements coming at the expense of interpretability should come as no surprise. Historically, Elo was developed with simplicity in mind with ratings often being calculated with limited computational resources. Especially with Elo being introduced as a new system, the need for it to be understood by players and organisers was paramount.

However fast forward to 2025, and while this simplicity may have been significant in promoting worldwide adoption of the system, we can now identify that this may have been prioritised over optimising key criteria like predictive accuracy and stability of ratings. With the now widespread availability of technology, tools and platforms to compute more complex rating systems, in addition to the findings of this section, we argue that now is the time to prioritise predictive accuracy and stability over the interpretability of the system.

Improved predictive accuracy ensures ratings reflect player strength more precisely, leading to better match predictions and increased trust from players, organisations and fans. Improved stability also ensures greater consistency in ratings over time. We infer from our data that, if interpretability is de-prioritised, the Elo system can indeed be improved through the introduction of a drawing propensity parameter or dynamic K-factors. While this demonstrates the clear potential for improving the Elo system, it is important to recognise the assumptions underlying these findings and our research’s limitations.

Our major limitation is in the quality of the data. There are many potential biases that may appear in our Lichess dataset, including a lack of games played using classical time controls and the representativeness of games played online. In particular we have assumed that our online dataset from November 2024 is representative of general chess play.

We also note that, due to dataset limitations, we excluded an analysis of scalability of the Elo system and Models A, B and C in this section; however we leave for further research to investigate how well these systems generalise across larger datasets with diverse player ratings and time controls.

We also leave for further research to investigate if additional models, that combine the improvements identified, can lead to breakthroughs that further improve on Models B and C from this section.

6 Applying the Elo Rating System to Football

6.1 Introduction

The Elo rating system was developed to rate chess players, and it has been used to rate the performance of teams in this game and other games such as football. In other words, the system is a new and improved ranking system that changes the rating value of the team each time it is changed depending on the result of the match and the competitor's strength. The goal of this paper is to find out whether the Elo system can be used in football and, mainly, in predicting future EPL positions. This paper will use all EPL results from 1993 to the present to assess the Elo system's accuracy in forecasting league outcomes and its applicability to rank teams' strengths and limitations. The Elo system is considered a more sophisticated and suitable approach than the points and goal differences methods. This means that it is helpful in football match analysis since it can be used to forecast the team's performance, given the team's and opponents' performance. Therefore, using the Elo system on the EPL dataset will make it easier to determine how well the ratings correspond with the league's final standings and the forecasts' efficiency.

6.2 Motivation and Relevance

The reasons for using the Elo system in football are evident in the ability of the system to predict future matches based on past performances, hence the system's importance to managers, analysts, and supporters in assessing the teams' capabilities. The current system of awarding points according to game results does not consider factors such as opposition strength, season scheduling, or short-term fluctuations in team form. In contrast, the Elo system recalibrates ratings after every match, thus incorporating dynamic factors like consistency in winning or losing streaks. Recent works highlight how factors of participation, performance variation, and competitive domains can significantly shape rankings and skill assessments [15–18].

6.3 The Elo Rating System and Football

The Elo rating system is based on the premise that the difference between two teams' ratings can determine the probability of one team beating the other. As we saw previously in Section 2.2, the formula for updating the Elo ratings is:

$$\text{New Elo} = \text{Current Elo} + K \times (\text{Actual Score} - \text{Expected Score}),$$

where:

- K is a constant that controls the sensitivity of rating changes (set to 10 as this produced the best results due to the low volatility of footballing performance).

- The match outcome determines the Actual Score: a win = 1, draw = 0.5, and loss = 0.
- Expected Score is the predicted result based on the difference in ratings between the two teams [41, 45]:

$$\text{Expected Score (home)} = \frac{1}{1 + 10^{\frac{(\text{AwayElo} - \text{HomeElo})}{400}}}.$$

It allows the teams to gain more points than expected by beating stronger teams or to lose fewer points than expected by drawing with teams they would have been expected to lose to. On the other hand, it punishes the teams that perform poorly, especially when they lose to weaker teams. However, the Elo system has its benefits when used in football since it can adapt to changes in a team's performance during the season. For example, a mid-tier team would have a higher Elo score after they achieve a series of victories, which means that their expected performance in the next games will be higher. On the other hand, teams that lose their games consecutively will see their ratings decrease, indicating their poor performance in the league.

6.4 Methodology: Python Implementation for the EPL

To assess how well the Elo rating system performs in predicting football results, we developed a Python script that processes all EPL results from 1993 to 2025. This section outlines the main steps of that script, ensuring our approach is clear before presenting the analysis.

6.4.1 Elo Initialization and Seasonal Loop

We maintained two dictionaries:

- `prev_season_Elos`: Elo ratings carried over from the end of the previous season;
- `alltime_Elos`: a record of each team's most recent Elo rating across all prior seasons.

When encountering a new season:

- If a team's Elo exists in `prev_season_Elos`, we use that.
- Else if the team has a rating in `alltime_Elos`, it inherits that rating.
- If the team is *fully new*, it inherits the lowest rating in the league for that coming season.
- For the very first season the initial Elo ratings are defined by the final league positions of each team in the previous season.

6.4.2 Match-by-Match Updates

Within each season, we iterate over matches in ascending date order. For each match:

1. **Home advantage:** We add +50 Elo points to the home team to compute the expected probability. This is similar to the way in which white are assigned additional rating in chess [41]
2. **Expected Score:**

$$p_H = \frac{1}{1 + 10^{\left(\frac{r_A - (r_H + 50)}{400}\right)}}.$$

3. **Actual Outcome:** $y_H = 1.0$ if the home team won, 0.5 if a draw, or 0.0 if the away team won.
4. **Update Elo:**

$$r'_H = r_H + K(y_H - p_H), \quad r'_A = r_A + K((1 - y_H) - (1 - p_H)).$$

The 50-point advantage is subtracted back after the update so only the pre-match calculation is biased.

5. **Points Tally:** We record 3, 1, or 0 points in a separate dictionary to compare at season end.
6. **Match-Level MAE:** We can capture $|y_H - p_H|$ to evaluate short-term predictive accuracy.

At season's end, these final ratings become `prev_season_Elos` for the next season, and we also update `alltime_Elos` for each team.

6.4.3 Performance Metrics

Spearman's Rank Correlation Coefficient To evaluate how closely Elo-based ratings align with actual points (or final standings), we employ *Spearman's rank correlation coefficient* (ρ) [46, 47]. Suppose we have n teams, each with Elo rating r_i and actual performance measure p_i . We define $\text{rank}(r_i)$ and $\text{rank}(p_i)$ as the respective ranks of team i (1 for highest, n for lowest). Then let:

$$d_i = \text{rank}(r_i) - \text{rank}(p_i).$$

Spearman's ρ is given by:

$$\rho = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}.$$

A value of ρ close to 1 indicates strong agreement between Elo ranks and actual ranks, while a value near 0 suggests little correlation.

MAE (rank difference).

$$\text{MAE}_{\text{rank}} = \frac{1}{n} \sum_{i=1}^n |\text{rank}(r_i) - \text{rank}(p_i)|,$$

showing how many places, on average, each Elo-based rank differs from the actual final table rank.

Match-level MAE. We further measure Elo’s short-term predictive accuracy with

$$\text{MAE}_{\text{match}} = \frac{1}{N} \sum_{j=1}^N |y_{H_j} - p_{H_j}|,$$

where p_{H_j} is Elo’s predicted win probability (home side), and $y_{H_j} \in \{1, 0.5, 0\}$ is the actual outcome from the home perspective, over N matches in a season (or across all seasons).

6.4.4 Overall Results and Summaries

After processing every match in every season, we:

1. Compute the final rating for each team in the last season, compare it to the *cumulative* points from 1993–2025 (yielding an overall Spearman correlation and rank-based MAE).
2. Aggregate all match-level errors from each season to produce an *overall match-level MAE* across the entire period.

6.5 Season-by-Season and Overall Results

In addition to comparing final Elo standings with actual league positions, we generate bar charts (Figures 10, 11, and 12) for each season’s metrics:

- Spearman correlation: Elo vs. final table points
- MAE (rank difference): average absolute difference between a team’s Elo rank and its actual rank
- Match-level MAE: short-term forecasting accuracy on a per-game basis

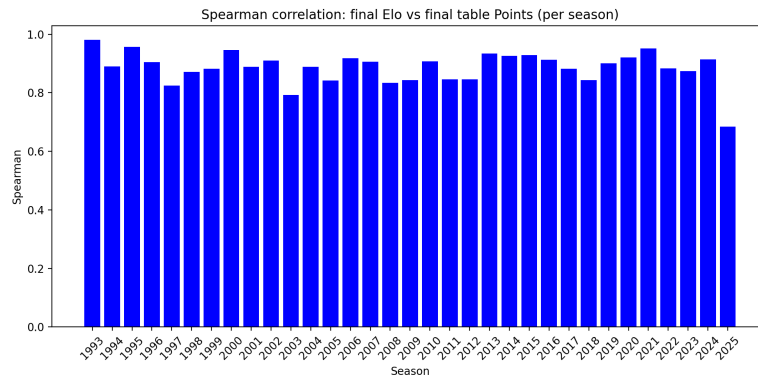


Figure 10: Spearman correlation of final Elo vs. final table points, per EPL season.

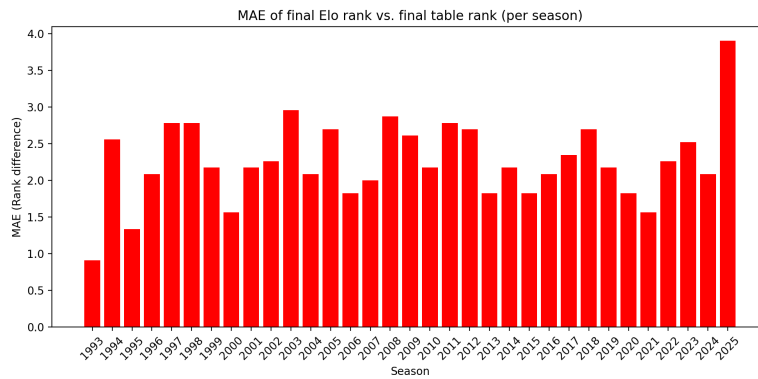


Figure 11: Rank-based MAE per season (average absolute difference between Elo rank and final table rank).

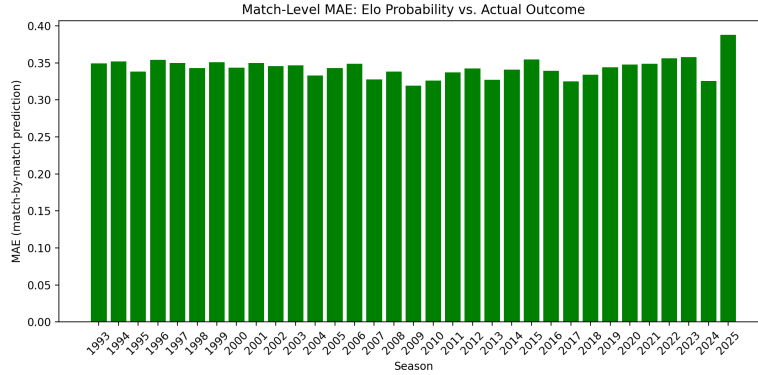


Figure 12: Match-level MAE per season, illustrating Elo’s per-game predictive error (home side).

Overall Printout and Values. Finally, the script produces a single correlation of final Elo (in 2025) vs. total historical points (1993–2025), plus the rank-based MAE on those all-time standings, and a combined match-level MAE across all fixtures. The result is:

```
Overall final-standings results across the entire dataset:
Spearman correlation (final) = 0.675
MAE (rank difference, final) = 3.800
Overall match-level MAE (prediction vs. actual) = 0.342
```

Hence:

- **Spearman = 0.675:** a moderate correlation between Elo’s final snapshot vs. all-time points.
- **Rank MAE = 3.800:** on average, about four ranks off from the total historical standings.
- **Match-level MAE = 0.342:** day-to-day outcome predictions deviate around 34% from the real results.

6.6 Limitations and Future Improvements

The Elo system thus has its drawbacks, yet it is quite efficient in predicting outcomes that are likely to occur within a season. The main disadvantage is the lack of consideration of factors affecting match outcomes, including injuries, red cards, or substitutions. For instance, this can be noted when a particular team loses one of its key players mid-season. This could be why the team’s performance may worsen, but the Elo system may take time to adjust the ratings for several matches. Similarly, changes in event factors such as managerial can lead to an immediate improvement in team performance, but the Elo system

may take time to adjust the team rating to the new level. As stated by [46], Elo ratings rely greatly on historical data of previous matches and do not consider other factors that may potentially impact the team’s performance.

Also, the Elo system assumes that the difference in the strength of the teams remains constant throughout the season. This is particularly the case in football because a team can go through a period of poor performance and then suddenly perform much better due to injuries, congestion and changes in form. For example, a team that has shifted its manager may improve its performance immediately, which may not be picked by the Elo system. This could significantly affect the disparity of the expected rankings if a team’s performance changes within a short period. In the same regard, [47] noted that the degree of fluctuation in the ratings depends on the frequency of the rating update. In future research, the Elo system could be improved by adding other factors such as injuries, changes in managers, or the level of difficulty of the games. This would make the system more responsive regarding the dynamic changes in the team performance and more accurate in predicting the game.

6.7 Conclusion

Using the Elo rating system, it is possible to provide a detailed and versatile evaluation of the football teams and their games. The Elo system is also preferable to a point system or a goal difference system as it can calculate a rating for the teams after a particular match. This has been used to see how effective the system is in determining the final standings of the teams in the EPL, whereby the majority of the teams have been seen to be in the same positions as their Elo ratings. It may be unable to account for a shock such as the 2015/2016 English Premier League champion in Leicester City or fluctuations in performance due to factors such as injuries or a change in the manager. Such limitations show that while the Elo system can be useful, it is insufficient to predict football results.

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